

Eugene (Yoogeun) Song

Date of birth: 27 November 1997 | **Nationality:** Korean (Republic of Korea)
Gender: Male (he/him/his) | **Location:** London, UK
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Education & Key Qualifications

Imperial College London <i>PhD Student in High-Energy Physics at Imperial</i>	2026 – 2030 London, UK
Imperial College London <i>Masters degree with the Distinction-awarded thesis project</i>	2024 – 2025 London, UK
University of Science and Technology <i>Graduate studies program in Astronomy and Space Science (Started in 2009 at 11 years of age: "Song Yoo-geun" project)</i>	2009 – 2018 Daejeon, Korea
National Institute for Lifelong Education <i>B.Sc. in Computer Science</i>	2006 – 2009 Seoul, Korea
Inha University <i>B.Sc. in Physics</i>	2006 – 2009 Incheon, Korea

Experience

PhD Researcher <i>Imperial College London</i>	2026 – Present London, UK
• I work on neutrino interactions, oscillations, detector analysis, and ML reconstruction techniques with Fermilab's (American) flagship collaborations, DUNE (Deep Underground Neutrino Experiment) and NOvA (NuMI Off-axis ν_e Appearance), under the supervision of Dr. Linda Cremonesi. • Systematics-aware ML reconstruction with Bayesian and MCMC inference for the DUNE Near Detector, and Near-Detector to Far-Detector constraint propagation for precision oscillation measurements. • A variety of physics analyses on NOvA, where methods are tested against real, systematics-limited data before being asked to carry a DUNE measurement.	
Research Scientist Intern <i>Singularity Quantum</i>	Aug 2026 – Present California, United States (Remote)
• I work alongside the research team across fluid dynamics, computational modelling, machine learning, software, quantum algorithms, and data analysis for cardiovascular computational fluid dynamics (CFD): blood-flow simulation, haemodynamic analysis, and the reconstruction problems underlying both. • I build the mathematical and computational models, simulations, quantum algorithms, software, papers, and technical reports that carry that work: from first formulation through to something reproducible. • I contribute to the research, development, testing, and documentation behind our cardiovascular products, principally FFR-CT and plaque analysis. • I design, train, and evaluate physics-informed machine-learning models. Foundational models where they earn their place; physics constraints where the data runs out. • I research tumour microcirculation, cancer-cell biology, and drug delivery, tracking the academic literature alongside the technologies, markets, and competitors moving in the same space. • I take on quantum computing research and the quantum engineering, analysis, and business work that sits around it.	

Graduate Researcher*Imperial College London*

Apr 2025 – Present

London, UK

- Architected and developed *AtriPINN*, an end-to-end physics-informed machine learning framework (Python/PyTorch) that reconstructs atrial fibrillation wavefronts in real time from 4×4 grid electrograms, leveraging integrated geometric frameworks to advance Afib mapping and treatment and elucidate the electrical mechanisms underlying irregular heartbeats.
- Engineered switchable physics back-ends (wave, eikonal and monodomain Aliev–Panfilov) within a single shared data $\rightarrow$ train $\rightarrow$ visualisation pipeline, with adaptive collocation, mixed-precision training and Adam $\rightarrow$ L-BFGS optimisation; shipped versioned releases v1–v11.2 with CLI tooling and reproducible configurations.
- Achieved ~ 78 ms end-to-end latency, ~ 1.6 mm RMS localisation error and ≥ 0.99 channel cross-correlation on clinical electrophysiology data from Hammersmith Hospital, against the 4–6 mm projection error typical of conventional electroanatomic mapping.
- Working with Prof. David Colling (Imperial HEP group) and Dr. Nick Linton (Imperial Department of Bioengineering), on clinical datasets from Hammersmith Hospital.
- MSc thesis on this work, *Physics-Informed Machine Learning for Real-Time Atrial Fibrillation Mapping from Grid Electrograms*, awarded the highest grade.

Quantitative Researcher*Independent Research Practice*

Oct 2025 – Aug 2026

London, UK

- Independent research on alpha generation under non-stationary market dynamics across futures, derivatives and equities, treating structural constraints (market impact, inventory, liquidity, latency) as explicit modelling anchors rather than post-hoc frictions.
- Studied agent adaptation and regime shifts, prioritising signals whose derivation survives a change in the data-generating process over in-sample backtest performance.
- Applied physics-inspired signal/noise separation, stochastic control (HJB), Monte Carlo methods and time-series modelling in Python.

UCL - MSSL visiting researcher*MSSL, University College London*

Jan 2023 – Mar 2024

London, UK

- Modelling Event Horizon Telescope images and multi-wavelength variability of Sgr A* using GRMHD simulations and GR radiative transfer post-processing, paper in prep.
- Working with Dr. Ziri Younsi and Prof. Kinwah Wu.

Independent Researcher*In collaboration with the National Astronomical Observatory of Japan*

Sep 2020 – Dec 2022

Remote

- During the COVID-19 pandemic, to maintain research momentum, despite pandemic-induced changes, I also collaborated remotely with Dr. Isao Okamoto (NAOJ) to develop modified Blandford–Znajek energy-extraction models for Kerr black holes (see arXiv:1904.11978 & 2401.12684).
- I also attended multiple conferences across Europe, including the key events in Spain, e.g. EAS 2022. Held discussions with various researchers in Europe while I was there, including Prof. Francisca Kemper, Dr. Frank M. Rieger, Dr. Benoît Cerutti, Prof. Božena Czerny, Dr. Oliver Porth, and Dr. Ziri Younsi.

National service*7th Engineer Brigade, VII Corps, Republic of Korea Army*

December 2018 – August 2020

Icheon, Korea

- Served as a soldier (final rank: Sergeant).
- Started as a Private, later promoted and finished as a Sergeant.
- As an Army driver, I was part of the brigade's transportation team.

Visiting Researcher*National Astronomical Observatory of Japan (NAOJ)*

Aug 2018 – Dec 2018

Tokyo, Japan

- Invited by Dr. Isao Okamoto for a four-month visiting research position in high-energy astrophysics, working on a modified Blandford–Znajek mechanism for electromagnetic energy extraction from rotating (Kerr) black holes.

- Developed a zero-angular-momentum-observer (ZAMO) treatment in which the force-free magnetosphere is divided at the null surface, showing that electromagnetic energy is self-extracted across it rather than drawn from the horizon at a distance.
- Reformulated the classical Blandford–Znajek process in terms of frame dragging, giving the energy transfer a surface with a clear physical meaning.
- Continued the collaboration remotely with Dr. Okamoto until the end of 2022, resulting in two arXiv preprints (arXiv:1904.11978 & arXiv:2401.12684).

Graduate/Postgraduate studies

2009 – 2018

University of Science and Technology

Daejeon, Korea

- Enrolled in the University of Science & Technology (UST) / Korea Astronomy and Space Science Institute (KASI) as a Master's student in 2009 (at age 11).
- Researching "High-Energy & Very-High-Energy Radiation from Black Hole Magnetospheres" under the guidance of Dr. Seok Jae Park at KASI.

Visiting Student

2017 – 2018

Academia Sinica, Institute of Astronomy and Astrophysics

Taipei, Taiwan

- Invited as a visiting researcher at the Academia Sinica Institute of Astronomy and Astrophysics from April 2017 to May 2018 to collaborate with Dr. Kouichi Hirotani, with whom I have published two peer-reviewed scientific papers.

Scientific Publications

1. **Yoogeun Song**, Hung-Yi Pu, Kouichi Hirotani, Satoki Matsushita, Albert K. H. Kong, Hsiang-Kuang Chang, "Enhanced gamma radiation towards the rotation axis from the immediate vicinity of extremely rotating black holes", *MNRAS Letters*, 471, 1 (2017), [10.1093/mnrasl/slx119](https://doi.org/10.1093/mnrasl/slx119)
2. Kouichi Hirotani, Hung-Yi Pu, Sabrina Outmani, Hsinhao Huang, Dawoon Kim, **Yoogeun Song**, Satoki Matsushita, Albert K. H. Kong, "High-energy and Very High Energy Emission from Stellar-mass Black Holes Moving in Gaseous Clouds", *ApJ*, 867, 2 (2018), [10.3847/1538-4357/aae47a](https://doi.org/10.3847/1538-4357/aae47a)
3. Isao Okamoto, **Yoogeun Song**, "Gravito-thermo-electrodynamic extraction of energy from Kerr black holes" (2019), [arXiv:1904.11978v1](https://arxiv.org/abs/1904.11978v1)
4. Isao Okamoto, Toshio Uchida, **Yoogeun Song**, "Energy self-extraction of a Kerr black hole with the force-free magnetosphere frame-dragged" (2024), [arXiv:2401.12684](https://arxiv.org/abs/2401.12684)
5. **Yoogeun Song**, Ziri Younsi, Yosuke Mizuno, Kinwah Wu, Christian M. Fromm, "Multi-wavelength variability of Sgr A* from GRMHD simulations with electron heating", in preparation (2024)

Notable Scientific Presentations and Conference Participation

• The 7th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy

Barcelona, Spain

July 4 – 8, 2022

• European Astronomical Society Annual Meeting 2022

Valencia, Spain

June 27 – July 1, 2022

• "Theoretical Models of the Black Hole Magnetospheres and Jets" in "Black Hole Astrophysics with VLBI: Past, Present, and Future."

NAOJ, Mitaka, Japan (1st author poster presentation)

March 27 – 29, 2017

- **Future Sky Surveys and Big Data**

KASI, Daejeon, Korea

April 25 - 29, 2016

- **3rd Korea - Japan Workshop on Dark Energy**

KASI, Daejeon, Korea

April 4 - 8, 2016

- **The Korean Astronomical Society 50th Anniversary Meeting 2015**

Seoul, Korea

April 15 - 17, 2015

Awards and Honors

Special Award

The National Assembly of the Republic of Korea

For undergraduate degree at **age 11**

February 2009

Child Prodigy Sponsorship Program

Samil PricewaterhouseCoopers (Korean branch of PwC)

24,000,000 KRW (~ \$18,500)

2014 - 2015

Super Child Prodigy Project

"Song Yoo-geun Project"

The Government of the Republic of Korea

100,000,000 KRW (~ \$77,000) (**KASI**)

2009 - 2015

Special (Pre - College) Scholarship & Monthly Subsidy 64,780,000 KRW (~ \$50,000)

Inha University

2006 - 2008

Professional Certification

Craftsman Information Processing certificate

Ministry of Science, ICT and Future Planning, Korea

2004 (at **age 6**)

Craftsman Information Equipment Operation

Ministry of Science, ICT and Future Planning, Korea

2004 (at **age 6**)

Computer Specialist in Spreadsheet & Database Level - 2 certificate

Ministry of Employment and Labor, Korea

2008 (at **age 10**)

Securities Education Certificate (Distinction)

Imperial College Investment Society

2025

Finance Accelerator

AmplifyME

2024

Scientific Public Engagement

- **500+ Korean and Japanese media appearances:** Various television, radio, and newspaper interviews. Since 2004, my achievements have been extensively documented in mainstream Korean media outlets, including the Kyunghyang Shinmun, Hankyoreh, Chosun Ilbo, Joongang Ilbo, Donga Ilbo, Seoul Broadcasting System (SBS), and Korean Broadcasting System (KBS). 2004 - 2023
- **SBS Special:** Documentary about my life in academia right before being enlisted in the Republic of Korea Army.
[YouTube](http://tinyurl.com/4atzjwx5) (<http://tinyurl.com/4atzjwx5>) October 2018
- **Seoul Digital Forum 2015:** Interview with Dr. Kip Thorne about the LIGO project.
[YouTube](http://tinyurl.com/mrxaz5n7) (<http://tinyurl.com/mrxaz5n7>) May 2015

- **EBS DocuPrime "Wonderful Science"**: Documentary titled "*The secret of the birth of the universe – Big Bang*," with me as presenter and narrator.
[Article](http://tinyurl.com/3z6sktun) (http://tinyurl.com/3z6sktun)
[Video](http://tinyurl.com/2p8w6em5) (http://tinyurl.com/2p8w6em5) May 2011
 (originally) November 2005
- **KBS "Screening Humanity"**: Two five-episode documentary series about my life in elementary and high schools as a prodigy.
[Article](http://tinyurl.com/yc5mxwef) (http://tinyurl.com/yc5mxwef) January and June 2005

Skills

Languages: English (*proficient*), Korean (*native*), French (*intermediate*), Italian (*A2*)

- **IELTS Academic:** Band 8.5 (L:9.0, R:8.5, W:7.0, S:8.5)
- **TOEFL iBT:** Score 106 (L:29, R:27, W:27, S:23)

Programming: LaTeX, Python, Fortran, C/C++, Mathematica, MATLAB, Julia, Bash

Other Interests

Musical Instruments: Guitar (Acoustic & Electric), Drums, Bass, Piano, Violin

Sports: Brazilian Jiu-Jitsu, Soccer, Golf, Tennis, Skiing/Snowboarding, Triathlon